

$$= 40 \text{ MJ kg}$$

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Answer: D

Since $v = \omega x_0 = 6.0 \text{ m s}^{-1}$ and $|a| = \omega^2 x_0 = 2.0 \text{ m s}^{-2}$,
 $\omega = |a| / v = 2.0 / 6.0 = 0.333 \text{ rad s}^{-1}$; $T = 2\pi / \omega = 18.9 \text{ s}$

$$x_0 = v / \omega = 6.0 / 0.333 = 18.0 \text{ m}$$

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Answer: A